Practical - 10

Secure Login/Logout with Sessions, Role-Based Access, Timeout, and Remember-Me Token

A . Theory

Develop secure login and logout functionality using PHP sessions. Verify password using password_verify(), regenerate session ID after login, restrict dashboard access, implement role-based redirection, session timeout, and optional secure remember-me token.

1. Are sessions started and destroyed securely?

Yes, sessions are started when the user logs in and destroyed when the user logs out.

2. Is `password_verify()` used correctly?

Yes, `password_verify()` is used to check whether the entered password matches the hashed password stored in the database.

3. Is unauthorized dashboard access blocked?

Yes, users must log in before accessing the dashboard. If they are not logged in, they are redirected to the login page.

4. Is session timeout implemented?

Yes, a session timeout is used to log out users automatically after a period of inactivity.

5. Are admin and student users redirected to appropriate dashboards?

Yes, after successful login, admin users are redirected to the admin dashboard, while student users are redirected to the student dashboard.

B. Implementation/ Output/Result

Here is my GitHub repo link the codes are here in repo:
https://github.com/dhyaani19/StudentHub-Portal.git

C.. Evidence of Work (Screenshots)

Figure 10.1 : Login Page

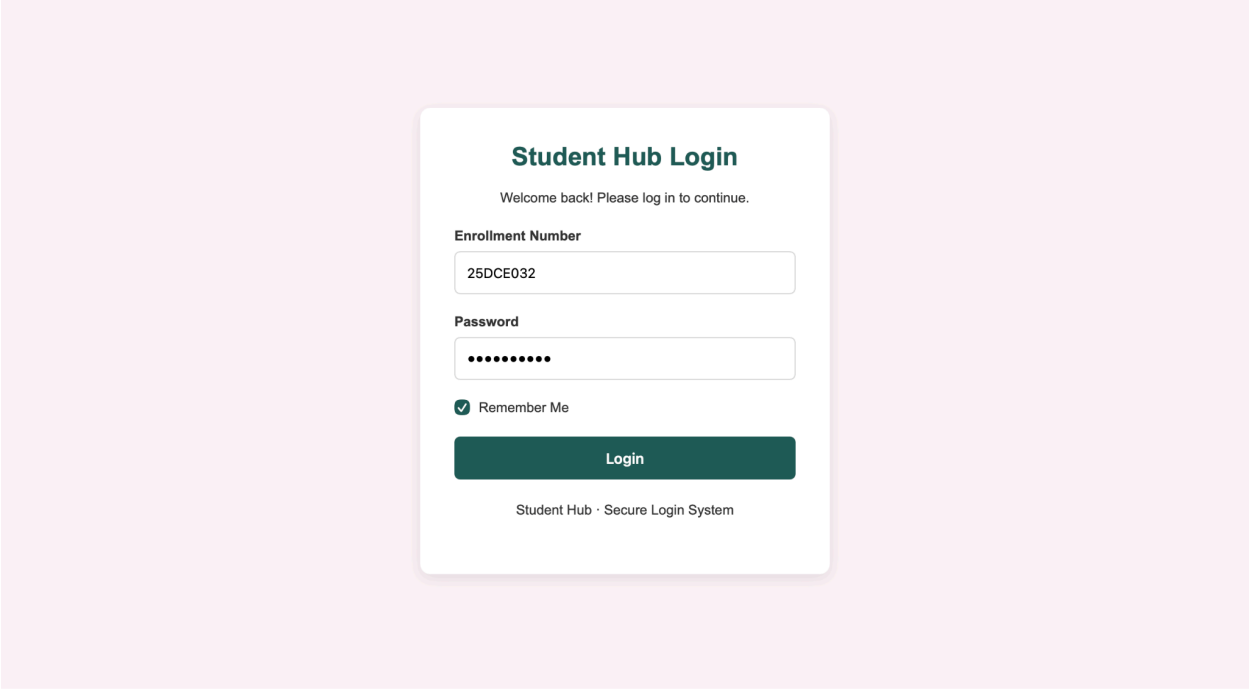


Figure 10.2 : Database Students

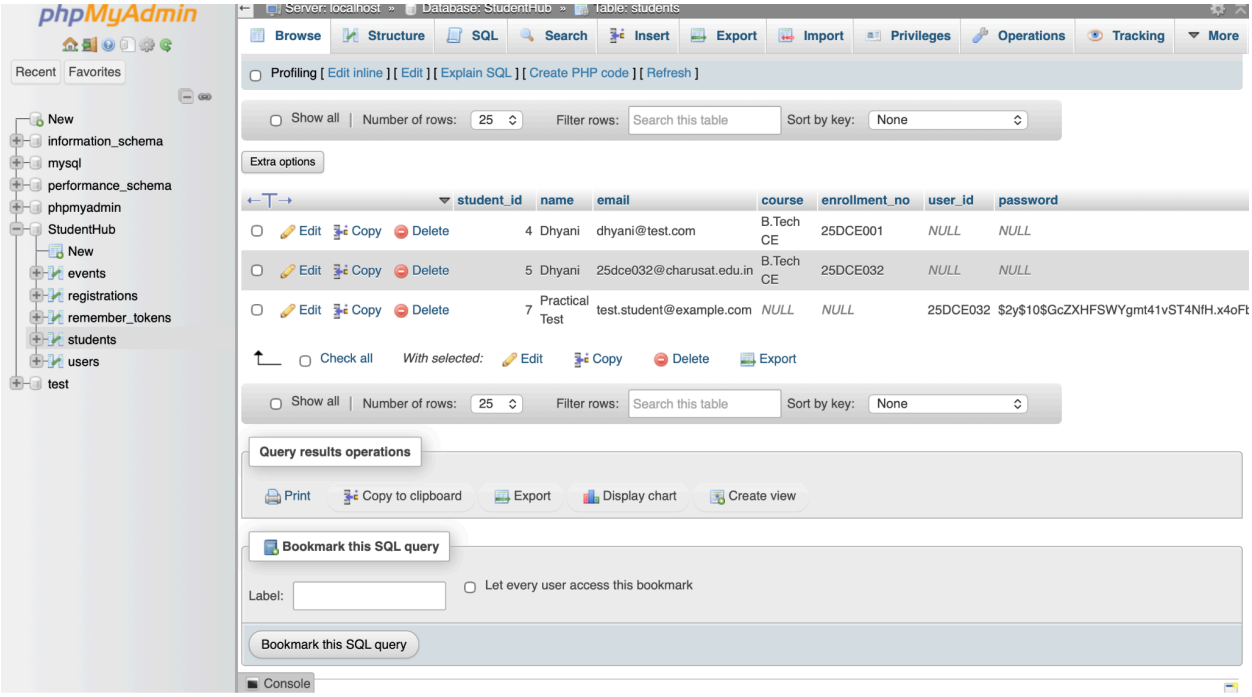
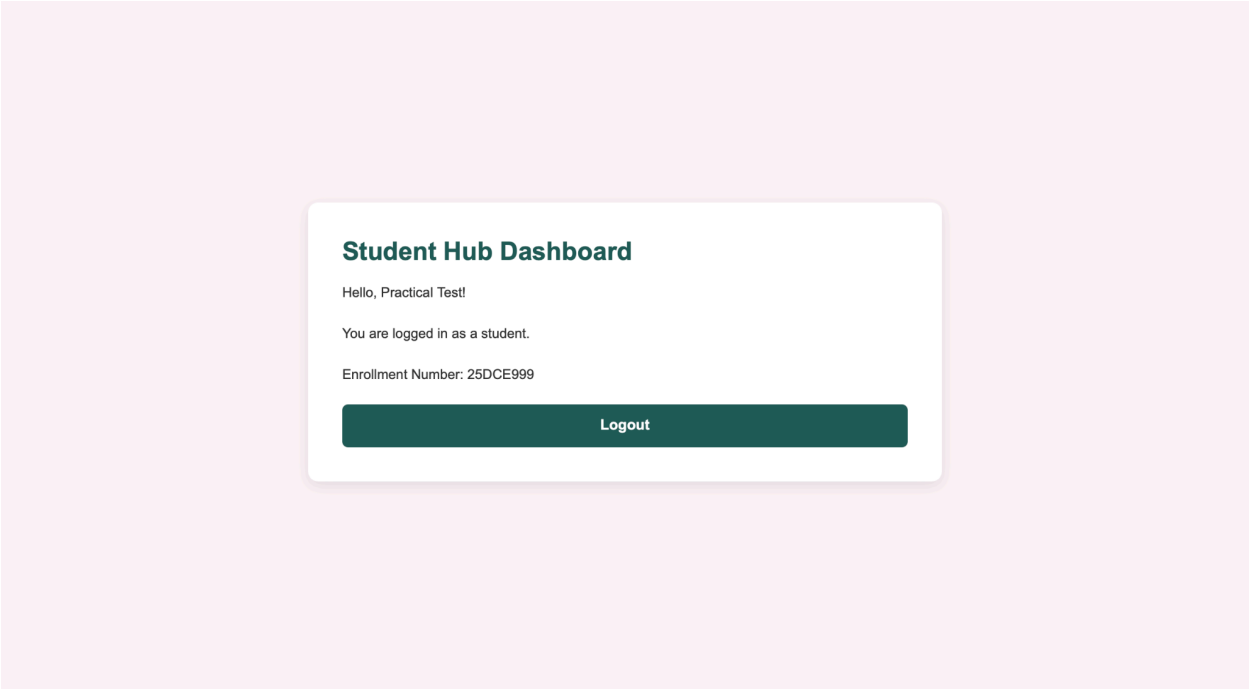


Figure 10.3 : Dashboard



D. GitHub Repository Link:

https://github.com/dhyaani19/StudentHub-Portal.git

E. Challenges Faced:

During this practical, I faced some difficulties in managing user sessions and making sure that only logged-in users could access the dashboard. I also had to understand how `password_verify()` checks passwords and why the session ID should be regenerated after login. Implementing session timeout and redirecting admin and student users to their respective dashboards was a little challenging. I learned to handle these situations by testing the login and logout process carefully.

F. Learning Outcomes:

Through this practical, I learned how to implement secure login and logout functionality using PHP sessions. I understood how to verify hashed passwords using `password_verify()`, regenerate session IDs after login, and restrict dashboard access to authorized users. I also learned how to implement session timeout and role-based redirection. Additionally, I understood how a remember-me token can help users stay logged in securely when implemented with proper security measures.

Faculty Remarks: